Hussain Singapuri

SENIOR AI ENGINEER (AGENTIC AI, TOOL CALLING AND LLM APPLICATIONS)

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PROFESSIONAL SUMMARY

AI Engineer with 3 years of professional software experience at a US-based medical-device company, building **production agentic AI** systems: **tool/function calling**, multi-step **multi-agent** orchestration with retries, timeouts and token budgets, **RAG** with **vector databases** (pgvector) and hybrid search, eval harnesses and output **guardrails**, on the **Claude API** and Gemini API. Strong **Python** backend skills (**FastAPI**, asyncio, **API design**, SQL), Docker and **CI/CD** on GitHub Actions. Leads technology as CTO at NeuroNetworks AI Ed. Claude Certified Developer.

CORE COMPETENCIES

Agentic AI & Multi-Agent Systems | Tool / Function Calling | Python, FastAPI & asyncio | API Design & Integration | RAG & Vector Search (pgvector) | Agent & LLM Evaluation | Claude & Gemini APIs | CI/CD, Docker & Git

WORK EXPERIENCE

NeuroNetworks AI Ed

2026 - Present

CTO & Principal Course Architect

Pune, India

- Lead technology for an Indian AI and tech education brand (Neuronetworks AI Ed Private Limited) offering programmes for students after Class 10
- Designing Neuromind Practical AI Lab, a browser-based IDE and sandbox with an AI tutor: phased spec, server sizing, engineering team plan and CAPEX/OPEX model
- Architected the curriculum: a 1-year AI Foundation course and five 3-year specialisations (AI/ML Engineering, Software Development, Data Science & AI, Product & UX / AI Design, Cybersecurity)
- Built and run the website neuronetwrks.com with a Python static-site generator deployed on Vercel, including programmatic SEO

US-based medical-device company

Mar 2023 - Sep 2026

Full Stack Developer

Remote

- Built **AI agents** for internal use
- Built and maintained full-stack web applications and several apps, including a **CRM**
- Performed penetration / security testing on 5+ of the company's mobile apps, all live on the Google Play Store

PROJECTS

multi-agent-research-crew

Open Source

Agentic AI: planner, parallel researcher, writer and verifier agents on a custom DAG orchestrator with retries, timeouts, token budgets and claim-level citation checking.

Python, asyncio, Claude

site-to-agent

Open Source

RAG agent: turns any website into a grounded chat agent with a crawler, hybrid BM25 + vector retrieval (pgvector) fused with RRF, cited answers, SSRF-hardened fetching and an eval harness for answer quality.

Next.js, TypeScript, pgvector

nanoserve Open Source

LLM inference engine from scratch in Python with an OpenAI-compatible streaming FastAPI API: paged KV cache, prefix caching, continuous batching, speculative decoding; 3.1x throughput over sequential decoding in benchmarks.

Python, PyTorch, FastAPI

PageForge Portfolio project

LLM grounding: AI drafts only from approved facts and the account's public pages, a deterministic validator flags unsupported numbers, a reviewer signs off claim by claim.

Next.js 16, React 19, Tailwind v4, Sanity, Supabase, Gemini, PostHog, HubSpot

contractor-ai-ops-dashboard Open Source

Multi-tenant analytics with signed event ingestion, funnel metrics, anomaly alerts and number-checked AI weekly summaries.

Next.js, TypeScript, Recharts, Postgres RLS

ai-lead-receptionist Open Source

AI chatbot for home-service businesses: missed-call text-back, SMS and web chat that qualifies leads and books jobs, with an owner dashboard.

Next.js, TypeScript, Twilio, Claude, Supabase

lead-intel-engine Open Source

Discovers, enriches and scores local-business leads with a robots-aware, rate-limited crawler and transparent scoring.

Python, FastAPI, SQLite

More projects

Orbital Surplus, Nightjar & Co., Personal portfolio with full-stack backend, Safey Inventory, TripSync, Lumen, Kilnwood Roasters, Cairnmoor, Lowlight Video Club

EDUCATION

Bachelor's, Computer Science (AI & Machine Learning) Vishwakarma Institute of Technology, Pune 2022-2025

Pune

CERTIFICATIONS

Claude Certified Developer	Anthropic, Foundations (CCDV-F)
Claude Certified Orchestrator	Anthropic, Foundations (CCAO-F)
Generative AI Leader	Google Cloud
Claude Code in Action	Anthropic
Claude Code 101	Anthropic

SKILLS

AI / LLM: Agentic AI, multi-agent orchestration, tool use / function calling, Claude API, Gemini API, RAG (hybrid BM25 + vector, RRF), embeddings, semantic search, evals, vector databases (pgvector), guardrails (prompt-injection defence, validators), LLM inference serving, Claude Code and AI coding agents

Back end: Python, FastAPI, asyncio, REST APIs, API design, Node.js, webhooks (HMAC signing), auth (scrypt, sessions, throttling)

Languages: Python, SQL, TypeScript, JavaScript, Dart, HTML, CSS

Data: PostgreSQL, Neon, Supabase (RLS, Edge Functions), PGlite, pgvector, SQLite, Sanity CMS

DevOps: Git, GitHub Actions, Docker, Vercel

Security: mobile app penetration testing, CSP and security headers, SSRF hardening, input validation

Automation & integrations: workflow automation, CRM integrations (HubSpot), PostHog, Twilio SMS, web crawling / scraping

Front end: React 19, Next.js, TypeScript, JavaScript, Tailwind CSS, shadcn/ui, TanStack Start / Query, GSAP, Lenis, three.js, WebGL, Canvas 2D, responsive and accessible UI, motion and interaction design

Mobile: Flutter, Riverpod, Firebase **Design:** UI/UX design, design systems, technical SEO, JSON-LD structured data, programmatic SEO